



DESIGN SPECIFICATION DOCUMENT

PROJECT: FLOAT REBALANCING – MOBILE
MONEY LIQUIDATION AT EQUITY -
DUUKA

Version 1.0.0

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REVISION HISTORY

Date	Version	Description	Revised By
19 th Feb, 2022	1.0.0	Initial Draft	Geoffrey Isaac Njoroge
9 th March 2022	1.0.1	Added MTN UG and Airtel UG Liquidation flowcharts	Geoffrey Isaac Njoroge

Sign Off required

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1. INTRODUCTION

Scope of this document

The purpose of this document is to provide a comprehensive specification for float rebalancing/ liquidation services for mobile money agents through float transfer to Telco agent bank account or float to cash withdrawal at the Equity Duuka Agent.

The scope covered in this document include:

- a) Float withdrawal / liquidation by Airtel Uganda Agent
- b) Float withdrawal service by MTN – Uganda Agent

2. DESIGN CONSTRAINTS, ASSUMPTIONS, DEPENDENCIES AND RISKS

Design Constraints and Assumptions

Assumptions

#	Description
1.	There exists interface to 3 rd party API providers
2.	An API gateway exists. All API calls must go through the API gateway
3	All dependent services are available
4	The business rules applicable for solution are available

Dependencies

- Integration to MTN and Airtel that will enable cash withdrawal/ liquidation
- Integration to Core Banking System (GAG, Finacle & MFS)
- High availability and good performance of the APIs
- Notification service for delivery of SMS alerts to Agents on cash withdrawals

Risks

1. Network connection failure
2. Network latency leading to increased timeouts and reversals.
3. Unavailability of the Telco provider leading to unavailable service
4. Delays or failure in SMS delivery

Logging

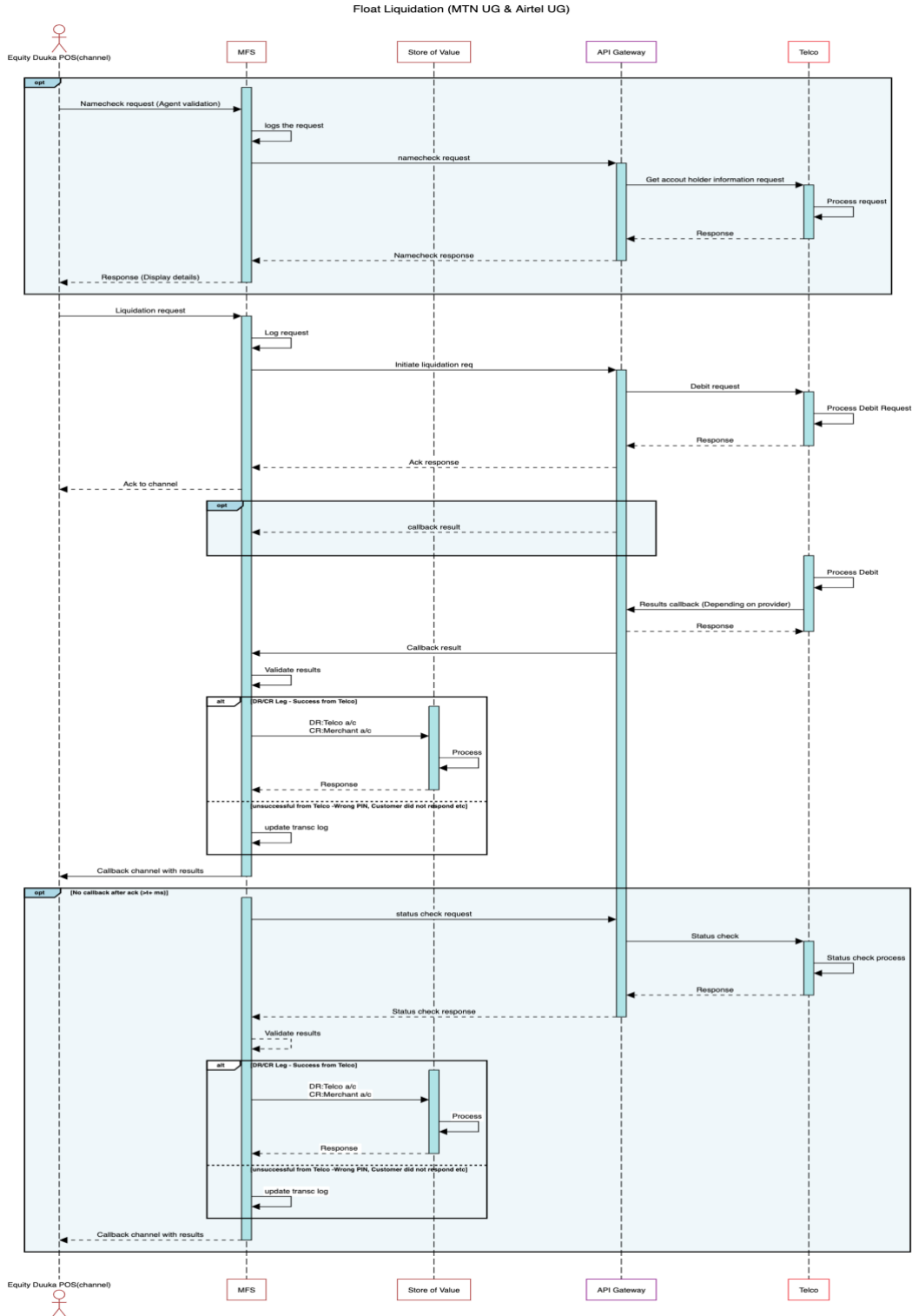
Standard Logging guidelines will be followed while implementing logging for the Equity Bank as follows;

1. The Input, Output and Error messages will be logged.
2. Any sensitive information (e.g. passwords, keys) will not be stored in the logs
3. An audit trail of privileged and user accounts activities will be maintained (including successful and failed login attempts).
4. All system components will have logging in place.
5. Log level will be controlled as a configuration

3. INTERFACE REQUIREMENTS

This service allows Telco Agents in Uganda to transfer their wallet float to their Equity bank account or withdraw it at an Equity Duuka Agent.

Below is the sequence flow for this service.





Below is the detailed description of the sequence flow:

- a) Telco Agent visits Equity Duuka agent and requests to transfer float from the Telco wallet to their bank account or withdraw as Cash
- b) Equity Duuka agents request the following information from the Telco agent
 - i. Telco agent code
 - ii. Amount to withdraw or transfer
 - iii. Bank account number and name to deposit to if account deposit
 - iv. For Airtel Uganda – Generated OTP code
 - v. Then Agent selects Telco provider – Either MTN Uganda or Airtel Uganda
- c) Information captured is entered on the Equity Agent App and submit to MFS
- d) First request to Telco is perform a Namecheck request to get Agent information.
- e) Upon successful namecheck request and confirmation of the request, initiate Liquidation request is submitted
- f) MFS validates the Equity Duuka agent liquidation request and if the payload is okay, a liquidation request is sent to the Telco via the API Gateway

MTN Uganda Agent Liquidation – Asynchronous call

- a) API gateway through VPN connection sends Debit request to MTN network
- b) MTN validates the request and responds with an acknowledgement
- c) MTN in the background initiates a pin validation request to the Telco agent to confirm transfer/withdrawal request
- d) Upon successful approval, MTN processes the request and sends a Callback result to complete transaction
- e) API Gateway via callback request receives the results and callback MFS that upon verification, updates the transaction log then calls Finacle/store of value to processes the Debit/Credit request
- f) Upon successful Liquidation a Callback to the Channel (Agent App) to inform Equity Duuka agent successful transaction to issue receipt and cash (if transaction request was float withdrawal)

Airtel Uganda Agent Liquidation – Synchronous call

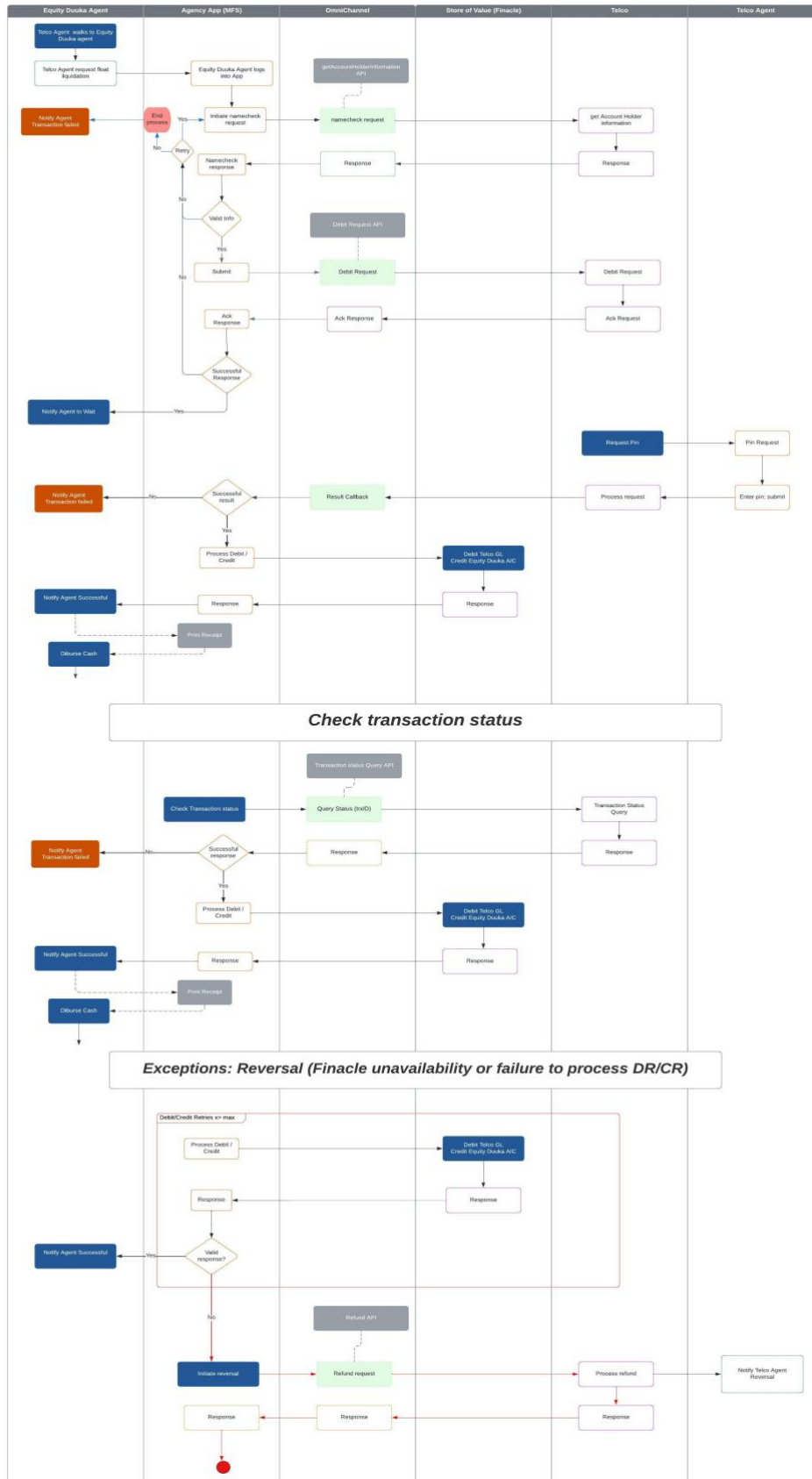
- a) Airtel Uganda integration does not support Namecheck validation
 - b) A Liquidation request is submitted to MFS to process request
 - c) MFS sends a cashback API call to Airtel via the API gateway
 - d) The API gateway sends a liquidation request and OTP confirmation to the Telco
- N/B:** Airtel agents first initiates a withdrawal request on Airtel menu to receive a One-Time-Pin (OTP) that is captured by the Equity Duuka agent during Liquidation request to the telco.
- a. Airtel verifies and validates the OTP submitted. (If OTP is valid and has not expired)
 - b. If the OTP verification and validation is successful, Airtel processes debit and credit
 - c. On successful processing, a response is sent to Liquidation service to complete Liquidation request
- e) MFS calls Finacle to process Debit/ Credit
 - f) A response is sent to Channel to generate transaction receipt
 - g) If transaction is cash withdrawal, Equity Duuka Agent issues cash equivalent to the transacted amount

Cash Withdrawal Vs Float transfer

- On cash withdrawal, Equity Duuka Agent float account is credited and the Telco GL account is Debited
- On transfer to Equity bank account, the Telco GL is debited and the Telco agent account is credited

MTN UGANDA LIQUIDATION WORK-FLOW

MTN Agent Float Liquidation Workflow



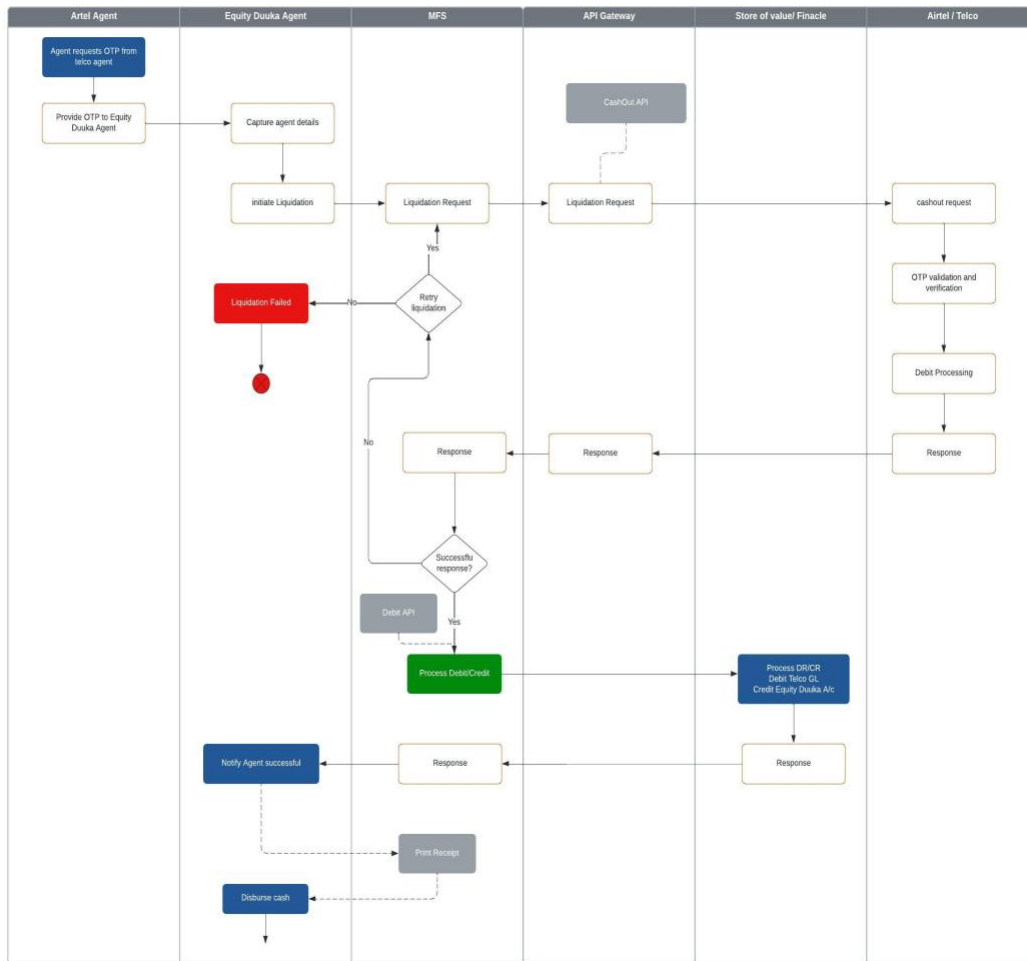


Detailed description:

- MTN agent walk to an Equity Duuka Agent and requests to withdraw float as cash
- Equity Duuka Agent logs into their Agent app and captures the following MTN agent details:
 - MTN agent number(ID) / MSISDN
 - Amount to withdraw
 - MTN agent full names
- Equity Duuka agent submits the request, MFS send a namecheck request to API Gateway and API gateway send a **getAccountHolderInformation** API request to MTN
- MTN responds with the Agent details which is presented to the Equity Duuka Agent
- The Equity Duuka agent can confirm the request or update and retry if the details do not match or cancel the request all together
- Upon submitting, **debit API** request is sent to MTN
- MTN responds with an acknowledgment
- MTN then sends a pin confirmation request to the MTN agent to enter pin
- After successful processing the response from MTN agent, Telco agent sends a callback
- MFS processes the callback and if successful result, MFS calls **Debit/Credit** API to finacle to Debit Telco GL and credit Equity Duuka Agent float account
- MFS then sends a callback to the channel (Equity Duuka Agent app) to disburse cash and print receipt
- In-case a callback is not receive in t-secs, MFS send a **transactionStatusQuery** to MTN

AIRTEL UGANDA LIQUIDATION WORK-FLOW

AirtelAgent Float Liquidation Workflow



Description:

- Airtel agent walks to Equity Duuka agent and requests to
 - Withdrawal float as cash
 - Transfer float to their Equity Account
- Airtel agent, using their telco menu generates an OTP that they provide to the Equity Duuka Agent
- Equity duuka agent log-into the app and captures the agent details, amount and account to deposit
- Equity Duuka agent initiates **cashout request**
- Airtel validates and verifies the OTP then perform cashout on their end.
- This call is synchronous, and therefore a response is sent back.
 - Successful response:
 - Cash withdrawal - Debit Telco GL and credit Equity Duuka float account
 - Float liquidation to account – Debit Telco GL and credit Airtel Agent Account
 - Failure response: Notify agent the transaction failed



INTERFACE DEFINITION

1) Transaction Request (MFS -> OMNI – API Gateway)

Channel Liquidation request to Business Engine

Interface ID	Liquidation
Version	1.0
Target Applications / Domains	Namecheck
Interface Type	REST/JSON
Mode	Asynchronous
Method	POST
Description	Equity Duuka Agent initiate liquidation of Agents float

Input Parameters

Field	Description	Format	Req.
providerID	Provider for the request	String	Y
channelID	Onboarded channel making the request	String	Y
requestID	Unique reference ID for the request	String	Y
serviced	Service ID	String	Y
narration	Request description	String	Y
liquidationType	Specify the type of liquidation either accountDeposit or cashWithdrawal	String	Y
countryId	Country ID for the request	String	Y
accountNumber	Account to credit in instances the liquidationType is accountDeposit	String	N
receiverName	The customer name	Date	Y
telcoAgentId	Agent code of the customer as provided for by the Telco	String	Y
currencyID	Charges & amount currency as setup for the service (ISO)	String	Y
Amount	Transaction amount	String	Y
Otp	Unique one-time pin generated by the Telco that authenticates the withdrawal		

Output Parameters (Acknowledgement to Channel)

Field	Description	Format	Req.
statusCode	Status code for the request	String	Y
referenceID	Unique reference ID for the request	String	Y
statusDescription	Description status for the request	String	Y



Sample Input

```
POST liquidation/v1 HTTP/1.1
Authorization: Bearer jeU5842ldkSgTODIHBaYoKQNPEbru6hCVin91Jqz
Content-Type: application/json
{
  "requestID": "retre990",
  "channelID": "channel1",
  "providerID": "airtel_ug", //mtn_uganda for MTN
  "serviceID": "liquidation",
  "narration": "Airtel Agent Liquidation at Equity Duuka",
  "liquidationType": "accountDeposit", //cashWithdrawal
  "countryId": "UG",
  "additionalParams": [
    {
      "key": "providerRef",
      "value": "MXte232323"
    },
    {
      "key": "accountNumber", //optional when doing account deposit but
      "value": "01234567891011"
    },
    {
      "key": "receiverName",
      "value": "Money receiver test"
    },
    {
      "key": "telcoAgentId", //Telco Agent number or MSISDN
      "value": "Airtel-6667845" //076300*****
    },
    {
      "key": "currencyId",
      "value": "UGX"
    },
    {
      "key": "amount",
      "value": 5000
    },
    {
      "key": "otp",
      "value": "123456"
    }
  ]
}
```



Sample Output

```
{
  "requestID": "retre990",
  "statusCode": "001",
  "statusDescription": "Request accepted successfully"
}
```

2) Callback sample (After successful transaction): OMNI -> MFS

Interface ID	Callback/v1
Version	1.0
Target Applications / Domains	Global telco engine
Interface Type	REST/JSON
Mode	synchronous
Method	POST
Description	This API will be used by global telco engine to call channels with the results

Parameter definition: Callback after transaction processing

Field	Description	Format	Req.
providerID	Provider for the request	String	Y
channelID	Onboarded channel making the request	String	Y
requestID	Unique reference ID for the request	String	Y
serviced	Service ID	String	Y
narration	Request description	String	Y
statusCode	Transaction status (Check mappings to see different status codes)	String	Y
statusDescription	Narration to the status code of the transaction	String	Y
referenceNumber	Transaction reference number. Will be used in cases of tracking transaction status	String	Y
additionalParameter	Additional data to the transaction	Object	Y

Sample input

```
POST Liquidation/callback/v1 HTTP/1.1
Content-Type: application/json
Authorization: Bearer MTQ0NjJkZmQ5OTM2NDE1ZTZjNGZmZjI3
{
  "requestID": "retre990",
  "channelID": "channell",
  "providerID": "airtel_ug, //mtn_uganda",
  "serviceID": "liquidation",
  "narration": "Airtel Agent Liquidation at Equity Duuka",
}
```



```
{
  "liquidationType": "accountDeposit", //cashWithdrawal
  "countryId": "UG",
  "internalRefNumber": "TPG_3939394",
  "statusCode": "00",
  "statusDescription": "Request fulfilled successfully",
  "additionalParams": [
    {
      "key": "providerRef",
      "value": "MXte232323"
    },
    {
      "key": "receiver name",
      "value": "Money receiver test"
    }
  ]
}
```

Sample output

```
{
  "reference": "ID892929",
  "statusCode": "00",
  "statusDescription": "Successfully updated"
}
```

1. Transaction status API

Interface ID	statusCheck
Version	1.0
Target Applications / Domains	Global telco engine
Interface Type	REST/JSON
Mode	Asynchronous
Method	POST
Description	This API will be used by channels to request for transaction status for a given request ID

sample input

```
{
  "requestID": "9283238",
  "originalRequestID": "73839489",
  "channelID": "oui",
  "providerID": "airtel_ug",
  "service": "liquidation"
}
```



sample output

```
{
  "requestID": "retre990",
  "channelID": "channel1",
  "providerID": "airtel_ug",
  "serviceID": "liquidation",
  "narration": "Airtel Agent Liquidation at Equity Duuka",
  "liquidationType": "accountDeposit", //cashWithdrawal
  "countryId": "UG",
  "internalRefNumber": "TPG_3939394",
  "statusCode": "001",
  "statusDescription": "Request fulfilled successfully",
  "additionalParams": [
    {
      "key": "providerRef",
      "value": "MXte232323"
    },
    {
      "key": "receiver name",
      "value": "Money receiver test"
    }
  ]
}
```

Response Codes

Global response codes within the bank network. each response code will have relevant description / narration

HTTP Response Code	Status Code	Description
200	00	Request processed successfully
400	89	Bad Request: Missing parameter; unsupported type or repeated parameter
400	98	Forbidden
401	95	Unauthorized: Invalid credentials
404	1	Resource Not found
500	99	Server error



4. TELCO API SPECIFICATIONS

AIRTEL API SPECIFICATION

This API specification will be used by the Equity bank to call Airtel network to:

- Validate / verify OTP submitted
- Process debit request: Debit Agent float account and credit the bank collection account

Request Sample

Field	Description	Req. (M/O)
PIN	PIN of Agent Line	M
MSISDN	MSISDN of Agent line without Country Code	M
MSISDN2	Customer MSISD without the country code	M
AMOUNT	Transaction Amount	M
USERNAME	User name for authentication	M
PASSWORD	Password for authentication	M
OTP	OTP initiated by Customer during cashout	M
TXNID	Transaction ID of Equity Bank	M

```
<COMMAND>
  <MSISDN>$MSISDN</MSISDN>
  <PIN>$PIN</PIN>
  <MSISDN2>$CUSTOMERNUMBER</MSISDN2>
  <PAYID>12</PAYID>
  <PROVIDER2>101</PROVIDER2>
  <PROVIDER>101</PROVIDER>
  <AMOUNT>$AMOUNT</AMOUNT>
  <OTP>$OTP</OTP>
  <TYPE>CASHOUTPAS</TYPE>
  <PAYID2>12</PAYID2>
  <TRID>$TXNID</TRID>
  <ISAMOUNTCHECKREQUIRED>Y</ISAMOUNTCHECKREQUIRED>
  <USERNAME>$USERNAME</USERNAME>
  <PASSWORD>$PASSWORD</PASSWORD>
</COMMAND>
```

Response

```
<COMMAND>
  <TYPE>CASHOUTPAS</TYPE>
  <TXNID></TXNID >
  <TXNSTATUS></TXNSTATUS>
  <MESSAGE></MESSAGE>
</COMMAND>
```



Response fields mapping

Field	Description	Req. (M/O)
TYPE	Service Response Type	M
MOBTXNID	AM Transaction id sent in response	M
TXNSTATUS	Transaction status if 200 success,01035 Pending, else fail	M
MESSAGE	Description message for the transaction status	M

MTN API SPECIFICATION

- Get account holder information (Namecheck API)

Sample request

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<ns0:getaccountholderinforequest
xmlns:ns0="http://www.ericsson.com/em/emm/provisioning/v1_1">
  <identity>ID: 256789999576/MSISDN</identity>
</ns0:getaccountholderinforequest>
```

Sample response

```
<?xml version="1.0" encoding="UTF-8"?>
<ns0:getaccountholderinforesponse
xmlns:ns0="http://www.ericsson.com/em/emm/provisioning/v1_1">
  <accountholderbasicinfo>
    <msisdn>256789999576</msisdn>
    <firstname>Emolut</firstname>
    <surname>Main</surname>
    <accountholderstatus>ACTIVE</accountholderstatus>
  </accountholderbasicinfo>
</ns0:getaccountholderinforesponse>
```

- Debit request – Initiate debit of customer telco float account

Sample request

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<ns2:debitrequest xmlns:ns2="http://www.ericsson.com/em/emm/financial/v1_1">
  <fromfri>FRI:26878642189/MSISDN</fromfri>
  <tofri>FRI:Equity-agent.sp1/USER</tofri>
  <amount>
    <amount>20</amount>
    <currency>UGX</currency>
  </amount>
</ns2:debitrequest>
```



```
</amount>
<externaltransactionid>123ABCFD4</externaltransactionid>
<referenceid>1234567890</referenceid>
</ns2:debitrequest>
```

Sample response (Acknowledgment)

```
<ns6:debitresponse xmlns:ns6="http://www.ericsson.com/em/emm/financial/v1_1"
xmlns:fic="http://www.ericsson.com/em/emm/financial/v1_0/common" xmlns:ns4="http://
/www.ericsson.com/em/emm/coupons/v1_0/common"
xmlns:op="http://www.ericsson.com/em/emm/v1_0/common"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <transactionid>26610501</transactionid>
  <status>PENDING</status>
</ns6:debitresponse>
```

- Transaction completed callback

Sample result

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<ns0:debitcompletedrequest xmlns:ns0="http://www.ericsson.com/em/emm/financial/v1_1">
  <transactionid>2661050</transactionid>
  <externaltransactionid>445</externaltransactionid>
  <receiverinfo>
    <fri>FRI:Equity-agent.sp1/USER</fri>
  </receiverinfo>
  <status> SUCCESSFUL </status>
</ns0:debitcompletedrequest>
```

Sample acknowledgement

```
HTTP Response code: 200
```

- Query Transaction status

Sample Request

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<ns2:gettransactionstatusrequest
xmlns:ns2="http://www.ericsson.com/em/emm/financial/v1_0">
  <referenceid>900057716798</referenceid>
</ns2:gettransactionstatusrequest>
```



Sample success response

```
<?xml version="1.0" encoding="UTF-8"?>
<ns2:gettransactionstatusresponse
xmlns:ns2="http://www.ericsson.com/em/emm/financial/v1_0">
  <referenceid>900057716798</referenceid>
  <status>SUCCESSFUL</status>
</ns2:gettransactionstatusresponse>
```

Sample failure response

```
<ns2:errorResponse xmlns:ns2="http://www.ericsson.com/lwac"
errorcode="TRANSACTION_NOT_FOUND"/>
```

Response Code Mappings

Value	Description
COMPLETED	Operation completed.
CREATED	Operation created.
EXPIRED	Operation expired.
FAILED	Operation failed.
PENDING	Operation pending.



5. SECURITY ARCHITECTURE

- Use of private and public keys for encrypting and decrypting key fields. This will be applied on selected fields in the api for non-repudiation
- Mutual SSL will be implemented where possible. Minimum requirement is to have one way SSL for transport encryption
- **Oauth 2.0 authentication will be used for APIs**
- **IP whitelisting**
- **IPSEC for VPN connections**
- API credentials will be stored in an encrypted format in the database (configurations)
- Database access will be restricted to the application user and application IP addresses
- All database activity will be monitored using existing monitoring mechanisms



6. RESOURCES REQUIRED

Developers from both telco and bank side are required to build required APIs and backend for this solution. QA resources will be required for testing. Production teams will also be involved to understand how the solution works

7. SERVICES SLA

Response times between telco systems and Equity bank systems should be within 2 seconds on average. Efforts will be made to ensure response times are lower than the average

8. PROCESS CHANGES

The proposed changes are new and therefore Equity Duuka agents will be trained on how to perform the float rebalancing / Liquidation. Agents will also need to be familiar with reconciliations as there will be a new transaction process (Liquidation)

9. IMPACT ON EXISTING SERVICES

MFS and omni will have increased traffic as these are new integrations. Omni has HA implemented. Monitoring of the existing services will continue to identify any additional capacity required

10. SERVICE CATALOGUE CHANGES

- No new services will be introduced

11. TRAINING REQUIREMENTS

Equity Duuka Agents will be trained on how to initial and transact a Liquidation request by the Telco agent

12. REFERENCED DOCUMENTS

Document	Attachment	Comments
BRD Document	 BRD float rebalancing- MM liq	
Airtel Cash-out API Spec document	 Airtel Integrated Cashout- StandardP	
MTN Liquidation API spec document	 MTN Uganda Agent Float Liquidations a	
Business case	 Business case- Float rebalancing- MM flc	